

UNIVERSITY OF GHANA
OMIS 308: Predictive Analytics

**Predictive Analytics for Global Real Estate Investment Strategy:
A Comparative Study of Accra, Mumbai, and London for ED
Happiness Estates**

Case Study Prepared for: Miss Osei Abrefi Akoto, CEO, ED Happiness Estates

Prepared by: MTN

Executive Summary

ED Happiness Estates has secured USD 500 million in venture capital funding to develop ultra-modern brutalist residential and commercial buildings for rental and sale in the world's busiest and most profitable cities. This report responds to Miss Osei Abrefi Akoto's request for data-driven guidance on where to deploy this capital. Our team selected three global cities spanning the full spectrum of economic development required by the assignment brief: Accra, Ghana (a developing economy and ED Happiness Estates' home region in West Africa), Mumbai, India (a large emerging-market metropolis), and London, United Kingdom (a mature, advanced-economy global financial capital).

For each city we assembled a real estate transaction or listing dataset, cleaned and engineered the data, and built at least two regression models and one to two time-series forecasting models to explain and forecast residential property prices. Across the three cities we estimated a combined total of six regression models (Linear Regression and a regularised counterpart, either Lasso or Ridge, for each city) and five time-series models (Simple Moving Average for all three cities, plus ARIMA for Accra and Mumbai and Holt's Exponential Smoothing for London), comfortably exceeding the assignment's minimum requirement of four predictive models.

The headline finding is that the three markets sit at very different points on the risk-predictability spectrum. London's price behaviour is the most explainable from observable data (Linear Regression $R^2 = 0.885$; Ridge $R^2 = 0.885$) and the most forecastable over time (Holt's Exponential Smoothing MAPE = 5.08%), consistent with its maturity, liquidity, and data richness. Mumbai occupies the middle ground, with a strong regression fit ($R^2 = 0.826$) driven by its very large sample size (over 45,000 listings) but a non-stationary, harder-to-forecast time series. Accra shows the weakest, though still meaningful, regression fit ($R^2 = 0.62$) and a monthly price series that behaves close to random noise around its mean once trend is accounted for, reflecting a thinner, less mature, and more informally-priced market.

We recommend a three-city, risk-tiered capital allocation strategy: anchor a large, long-duration rental-focused tranche in London for stable, forecastable cash flow; deploy a growth-oriented, sales-and-rental-mixed tranche in Mumbai to capture emerging-market appreciation at scale; and enter Accra with a smaller, phased tranche targeting high-value localities close to the Central Business District, using it as ED Happiness Estates' flagship regional market in West Africa. Five detailed, data-driven recommendations, along with the supporting regression

and time-series evidence, risk considerations, and positioning guidance for each city, are presented in the Recommendations section of this report.

Table of Contents

1. Introduction.....	5
2. City Selection and Justification	6
3. Data Sources	8
4. Methodology	9
5. Analysis and Results: Accra, Ghana (Developing Economy)	11
6. Analysis and Results: Mumbai, India (Emerging Economy)	15
7. Analysis and Results: London, United Kingdom (Advanced Economy)	18
8. Comparative Discussion	22
9. Risks and Limitations	23
10. Recommendations for ED Happiness Estates.....	25
11. Conclusion	27
References.....	28

Introduction

Real estate investment decisions increasingly rely on data-driven evidence rather than intuition alone, particularly when large sums of capital and multiple international jurisdictions are involved. ED Happiness Estates, a rapidly growing West African real estate firm led by CEO Miss Osei Abrefi Akoto, has secured USD 500 million in funding from 35 venture capitalists and other global investors for the development of ultra-modern brutalist architecture buildings intended for both rental and sale. Overwhelmed by investor expectations and uncertain about the most strategic locations for this capital, Miss Akoto commissioned this analysis to identify which city or cities offer the strongest, most defensible investment case.

This report applies predictive analytics techniques, regression modelling and time-series forecasting, to real-world housing market data from three cities representing different stages of economic development: Accra (developing economy), Mumbai (emerging economy), and London (advanced/developed economy). The objective is twofold: first, to identify and quantify the factors that most strongly drive housing prices and sales in each city; and second, to use these models, together with forecasts of future price movement, to produce a ranked, well-justified set of investment recommendations for ED Happiness Estates.

Objectives

- Select and justify three global cities spanning developed, emerging, and developing economies as candidate investment markets.
- Collect, clean, and engineer real estate and socio-economic data for each city from multiple reputable sources.
- Build and evaluate at least two regression models and two time-series forecasting models to explain and forecast housing prices.
- Interpret model results to identify the key determinants of price in each market.
- Translate the analysis into at least five well-justified, data-driven investment recommendations for ED Happiness Estates.

City Selection and Justification

In line with the assignment brief, our team selected three major global cities, ensuring representation of one advanced/developed economy, one emerging economy, and one

developing economy. The selected cities are Accra (Ghana), Mumbai (India), and London (United Kingdom).

Accra, Ghana: A Developing Economy

Accra is ED Happiness Estates' home market and the commercial capital of Ghana, West Africa's second-largest economy. The city has experienced rapid urbanisation and a growing middle class, driving demand for modern residential developments in localities such as East Legon, Cantonments, and Airport Residential Area. As a developing economy, Accra offers high growth potential and first-mover advantage for a brutalist-style development concept that is not yet widespread in the local market, but it also carries higher market and institutional risk (less mature title/legal infrastructure, thinner transaction volumes, and more informal price negotiation) than the other two cities. Including Accra also directly serves ED Happiness Estates' strategic interest in strengthening its position in its home region.

Mumbai, India: An Emerging Economy

Mumbai is India's financial capital and one of the most populous metropolitan areas in the world, sitting within an emerging economy characterised by strong GDP growth, a rapidly expanding middle class, and heavy investor interest in real estate as India's urban population continues to grow. Mumbai's real estate market spans a wide spectrum from affordable to ultra-luxury ("tier" segments in our dataset), giving ED Happiness Estates the flexibility to position brutalist developments at multiple price points. Mumbai's large transaction volumes and increasing metro/transit connectivity also support long-run capital appreciation and market size that a developing economy like Accra cannot yet match.

London, United Kingdom: An Advanced Economy

London is a mature, globally systemic financial centre with one of the world's most liquid, transparent, and well-documented residential property markets. It offers economic and political stability, strong legal protections for property investors, and consistently high demand from both domestic and international buyers and tenants. While price growth is typically slower and entry costs (£ per square metre) far higher than in Accra or Mumbai, London anchors the portfolio with predictability and downside protection, making it the natural "stable core" holding within a three-city, risk-diversified strategy.

Criterion	Accra	Mumbai	London
Market size	Small, high-growth	Very large, high-growth	Large, mature
Growth potential	High	High	Moderate
Investor interest	Rising, niche	Very high	Consistently high
Economic stability	Moderate	Moderate-High	High
Price predictability (our R ²)	0.62	0.83	0.88
Strategic role for ED Happiness Estates	Regional flagship / growth bet	Scale & diversification	Stable anchor / rental yield

Table 1. Summary comparison of the three selected cities.

Data Sources

Consistent with the requirement to draw on multiple reputable sources, our team collected city-level residential real estate transaction/listing data supplemented by relevant socio-economic and locational variables. Table 2 summarises the primary dataset used for each city.

City	Primary Dataset	Approx. Size	Key Variables
Accra	Accra residential property listings dataset (Kaggle-sourced, Greater Accra region)	4,320 listings (3,917 after filtering to Greater Accra)	Price, category, bedrooms, bathrooms, floor area, condition, furnishing, parking, latitude/longitude, locality, date
Mumbai	Mumbai property sales dataset (Kaggle-sourced)	50,000 listings (~45,200 after data-quality filtering)	Price (INR/USD), locality, region, market tier, carpet/built-up area, bedrooms (BHK), floor, building age, furnishing, parking, builder tier, metro proximity
London	UK Government / Kaggle "Housing in London" monthly borough dataset	13,549 monthly borough records (7,328 after filtering to valid boroughs, 1995–2020)	Average price, borough (area), houses sold, number of recorded crimes, date

Table 2. Primary data sources used in the analysis.

These primary transaction datasets were supplemented with contextual socio-economic reasoning drawn from publicly available macro-level sources recommended in the assignment brief, including the World Bank Database (GDP growth, urbanisation rates), Numbeo (comparative cost-of-living and rent indices), and national statistical/housing authority publications for Ghana, India, and the United Kingdom, which informed the city-selection justification in Section 2 and the qualitative risk discussion in Section 9. All datasets were geographically filtered to their relevant metropolitan area (Greater Accra region; Mumbai Metropolitan Region localities; Greater London boroughs) before analysis to ensure relevance to housing prices, rental rates, and market demand as required by the brief.

Methodology

All analysis was conducted in Python using pandas, NumPy, scikit-learn, statsmodels, and pmdarima, following a consistent analytical pipeline across the three cities: data understanding,

data cleaning and integration, outlier detection and handling, feature engineering, exploratory visualisation, regression modelling, and time-series forecasting.

Data Cleaning and Integration

- Checked and confirmed the absence (or extent) of missing values and duplicate records in each raw dataset.
- Filtered each dataset to the correct metropolitan boundary (e.g., excluding non-Greater-Accra regions; excluding London's own regional aggregate rows and boroughs with excessive missing data).
- Parsed date fields into proper datetime objects and derived year/month features to support both regression and time-series analysis.
- Removed logically inconsistent records (e.g., Mumbai listings where the floor number exceeded the building's total floors, or the construction year post-dated the listing year, resulting in 4,793 rows, or 9.6% of the dataset, being removed on this basis).
- Dropped target-leakage and redundant columns prior to modelling (e.g., Mumbai's price-per-square-foot and INR price columns, which are mathematically derived from the USD price target; London's non-borough regional aggregate rows).

Outlier Detection and Handling

Outliers were identified using the interquartile range (IQR) method. For Accra, price and floor-area outliers were capped at the IQR bounds rather than deleted, to preserve sample size in an already-moderate dataset. For Mumbai, outlier incidence was profiled by market tier (luxury 4.8%, mid-market 6.0%, premium 6.5%, affordable 6.6%) to confirm that extreme values reflected genuine luxury-segment pricing rather than data errors, before logically-inconsistent records were removed as described above.

Feature Engineering

- Accra: price per square metre; bedroom-to-bathroom ratio; Euclidean distance from the Accra Central Business District (computed from latitude/longitude and converted to kilometres); ordinal encoding of furnishing level and property condition; one-hot encoding of property category and locality (rare localities with fewer than 30 listings grouped into an "Other" category to control dimensionality).

- Mumbai: property age (listing year minus year built); numeric bedroom count (BHK) extracted from the property-type field; one-hot encoding of low-cardinality categorical variables (region, market tier, facing direction, furnishing, parking, builder tier, metro-distance type, metro line, redevelopment status); mean target encoding of high-cardinality categorical variables (locality, builder, metro station) to avoid an unmanageable number of dummy columns.
- London: year and month extracted from the date field; one-hot encoding of borough (area); missing houses-sold values imputed with the borough median.

Predictive Modelling Approach

For each city we built at least two regression models to explain cross-sectional price variation and at least one time-series model to forecast price movement over time, together satisfying the assignment's requirement of a minimum of four predictive models across the project. In total, six regression models and five time-series models were estimated across the three cities:

- Regression models: Ordinary Least Squares Linear Regression (baseline, interpretable coefficients) and a regularised counterpart, either Lasso for Accra or Ridge for Mumbai and London, chosen to test the baseline model's robustness to multicollinearity among the many one-hot-encoded location dummies and to guard against overfitting.
- Time-series models: Simple Moving Average (SMA, a naive but interpretable benchmark) for all three cities; ARIMA / seasonal ARIMA, selected automatically via stepwise AIC minimisation (auto_arima), for Accra and Mumbai; and Holt's Exponential Smoothing (trend-adjusted) for London, chosen after the SMA benchmark underperformed on London's clear long-run upward trend.

All regression features were standardised (z-scored) prior to model fitting, and each dataset was split 80/20 into training and test sets (random_state = 42) to obtain out-of-sample performance estimates. Model performance was evaluated using R^2 (variance explained), Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and, for time-series models, Mean Absolute Percentage Error (MAPE). Stationarity of each city's monthly price series was formally tested using the Augmented Dickey-Fuller (ADF) test prior to ARIMA

Analysis and Results: Accra, Ghana (Developing Economy)

Data Understanding and Preparation

The raw Accra dataset contained 4,320 property listings across 13 fields, with no missing values and no duplicate records. After filtering to the Greater Accra region, 3,917 listings remained. Average listed price was approximately GHS 13,039 (median GHS 8,000), with a right-skewed distribution and a maximum of GHS 59,500 prior to outlier treatment. After IQR-based capping of price and floor area, engineering of derived features (price per square metre, bedroom-to-bathroom ratio, distance from the CBD, sale year/month), and one-hot encoding of property category and locality, the regression-ready dataset expanded to 47 columns.

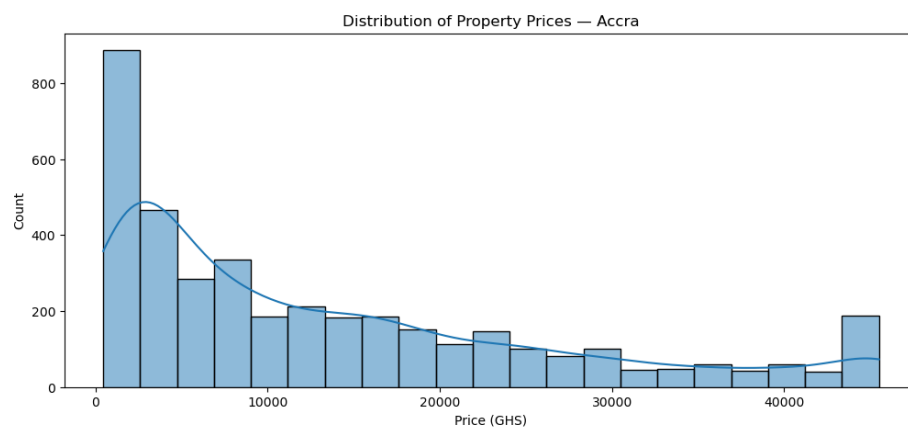


Figure 1. Distribution of property prices in Greater Accra (post outlier-capping).

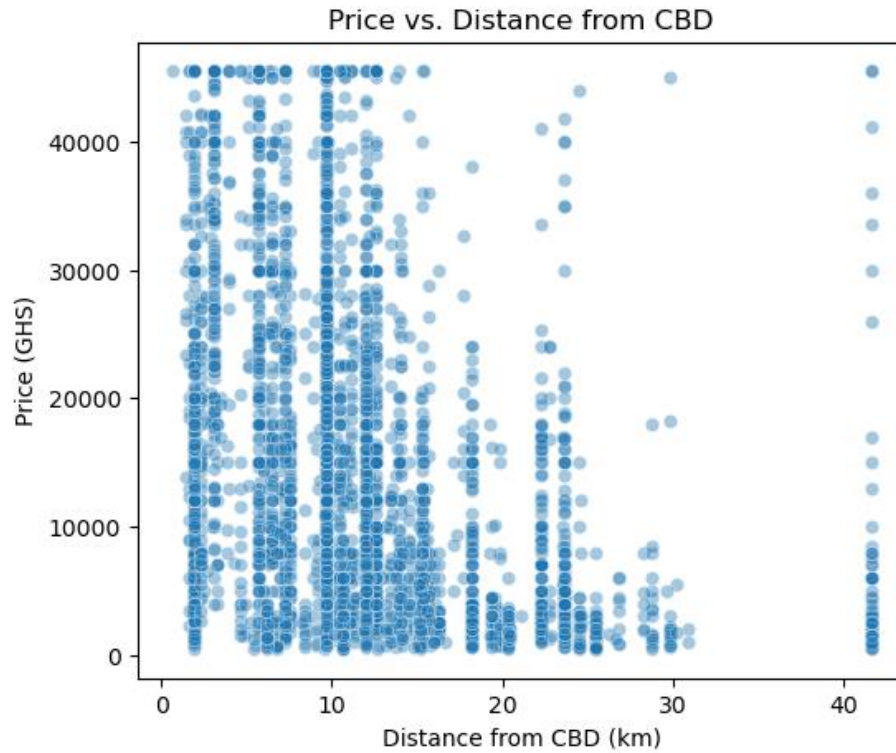


Figure 2. Relationship between property price and distance from the Accra Central Business District.

Figure 2 shows a visible, if noisy, negative relationship between distance from the CBD and price: properties closer to central Accra tend to command higher prices, consistent with the assumption that centrality is a valuable, scarce attribute in a developing-economy market with congested transport infrastructure.

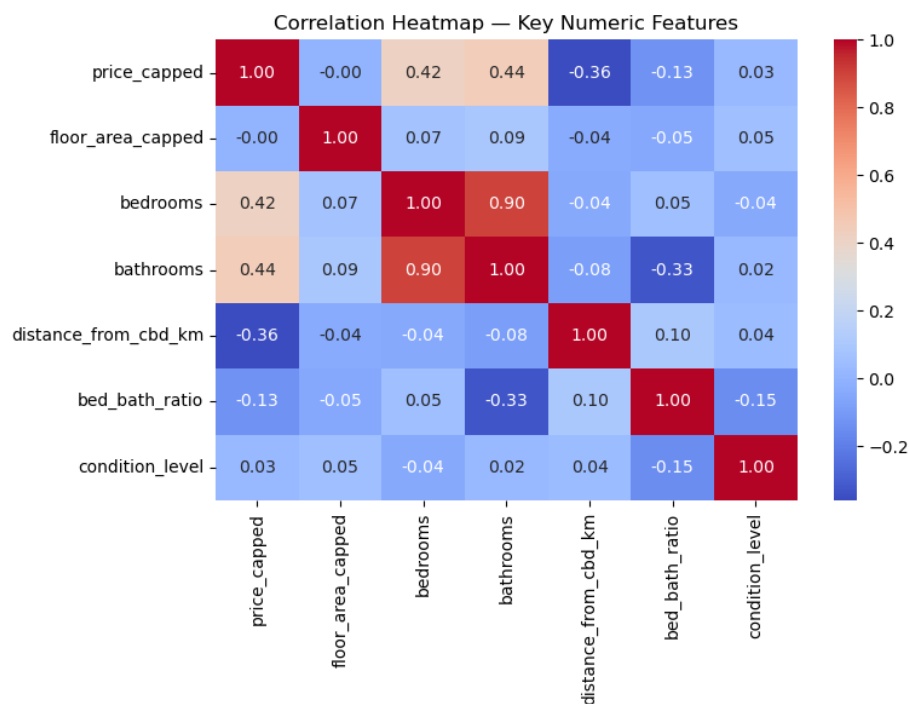


Figure 3. Correlation heatmap of key numeric features against capped property price, Accra.

Regression Modelling

A Linear Regression model was fitted on the standardised feature set (property characteristics, condition, furnishing, locality dummies, and engineered distance/time features) to predict capped property price. The model achieved an R^2 of 0.623 and a Mean Absolute Error of approximately GHS 6,029 on the held-out test set. A regularised Lasso regression (5-fold cross-validated, optimal $\alpha = 50$) produced an almost identical R^2 of 0.623, confirming that the Linear Regression result is stable and not an artefact of overfitting to the 47-feature space.

Model	R^2	MAE (GHS)
Linear Regression	0.623	6,029
Lasso Regression ($\alpha = 50$)	0.623	N/A

Table 3. Accra regression model performance (test set).

An R^2 of roughly 0.62 indicates that observable property characteristics explain a meaningful majority, but not the overwhelming bulk, of price variation in Accra, a pattern consistent with a developing real-estate market where informal negotiation, title/legal status, and off-market factors not captured in listings data still play a substantial role in final transaction prices.

Time-Series Forecasting

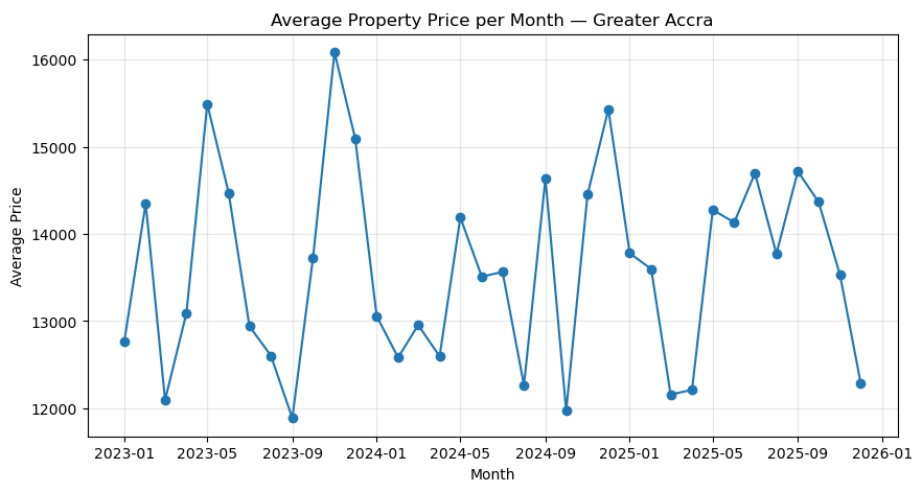


Figure 4. Average monthly property price, Greater Accra.

Monthly average prices were aggregated and forecast using a 3-month Simple Moving Average, which achieved a MAPE of 10.95% (MAE GHS 1,564; RMSE GHS 1,702) on the held-out test period. An Augmented Dickey-Fuller test on the monthly series strongly rejected the null hypothesis of non-stationarity ($p \approx 9.9 \times 10^{-6}$), indicating the series is already stationary in levels. A stepwise auto-ARIMA search (over the 36 monthly observations available) selected

ARIMA(0,0,0)(0,0,0)[12] with a constant term as the best-fitting specification (AIC = 610.1), effectively indicating that, once the mean level is accounted for, month-to-month price movements in this dataset are close to statistical noise, with no strong autoregressive or moving-average structure to exploit. This is consistent with Accra's smaller sample size and less mature price-discovery process, and implies that near-term price forecasts should be treated with wide confidence bands.

Model	MAE	RMSE	MAPE
3-Month SMA	GHS 1,564	GHS 1,702	10.95%
ARIMA(0,0,0)(0,0,0)[12]+c	N/A	N/A	AIC = 610.1

Table 4. Accra time-series model performance.

Analysis and Results: Mumbai, India (Emerging Economy)

Data Understanding and Preparation

The Mumbai dataset is substantially larger than the Accra dataset, comprising 50,000 property listings across 32 fields, including locality, market tier (luxury, premium, mid-market, affordable), carpet and built-up area, bedrooms, floor and total building floors, year built, furnishing, parking, builder tier, and metro-station proximity. The data contained no missing values or duplicate rows. Target-leakage columns (price per square foot and the INR-denominated price, both mathematically derived from the USD price target) were dropped, and a redundant area column was removed to reduce multicollinearity. Logical-consistency filtering removed 4,793 records (9.6%) where the listed floor exceeded total floors or the construction year post-dated the listing date.

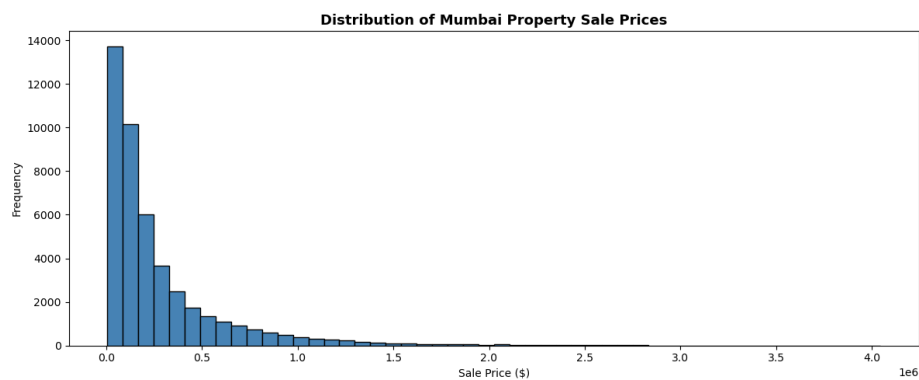


Figure 5. Distribution of Mumbai property sale prices (USD).

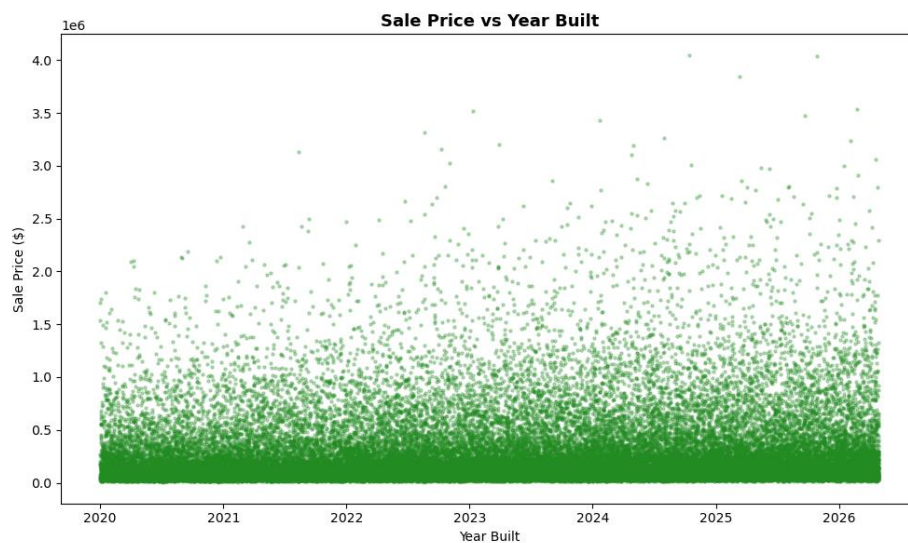


Figure 6. Mumbai sale price against listing date, showing overall price dispersion and upward drift over time.

Feature engineering derived property age (listing year minus year built) and a numeric bedroom count (BHK) from the property-type field. High-cardinality categorical variables, namely locality, builder, and metro station, were target-encoded (mean price per category) rather than one-hot encoded, keeping the feature space manageable (51 columns) while retaining the substantial locational price signal these variables carry; low-cardinality categoricals were one-hot encoded. The cleaned dataset was split into 36,165 training and 9,042 test observations.

Regression Modelling

Linear Regression achieved an R^2 of 0.826 and a Mean Absolute Error of approximately USD 81,769 on the test set. A 5-fold cross-validated Ridge regression (optimal $\alpha = 1.0$) produced a virtually identical R^2 of 0.826, again confirming the robustness of the Linear Regression fit. The considerably stronger R^2 than Accra's (0.826 vs 0.623) reflects both Mumbai's much larger sample size (over 45,000 usable listings) and the richness of its feature set, particularly market-tier segmentation and target-encoded locality/builder information, which capture fine-grained locational and quality premiums that the Accra dataset's simpler locality dummies cannot.

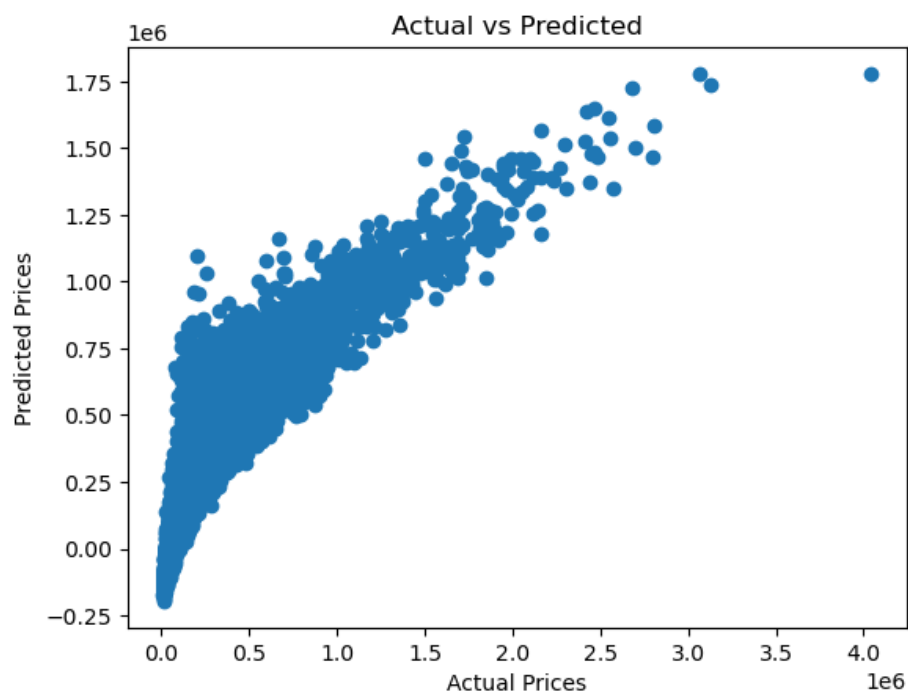


Figure 7. Actual versus predicted Mumbai sale prices (Linear Regression, test set).

Model	R^2	MAE (USD)
Linear Regression	0.826	81,769
Ridge Regression ($\alpha = 1.0$)	0.826	N/A

Table 5. Mumbai regression model performance (test set).

Time-Series Forecasting

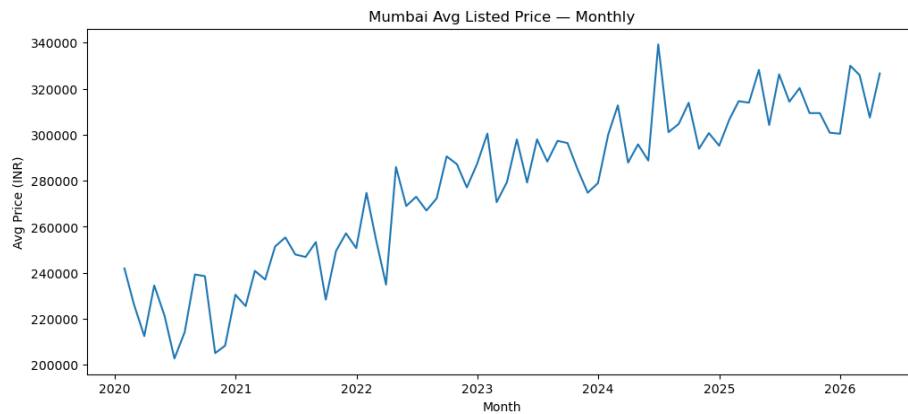


Figure 8. Average monthly Mumbai listing price.

An Augmented Dickey-Fuller test on the monthly average-price series failed to reject the null hypothesis of non-stationarity ($p = 0.073$), indicating the presence of a trend. After first-differencing, the series became strongly stationary (ADF $p \approx 9.5 \times 10^{-11}$), consistent with Mumbai exhibiting a persistent underlying price trend, plausible given India's sustained urban housing demand growth, rather than a level series that simply fluctuates around a fixed mean, as observed in Accra. A stepwise auto-ARIMA search on the differenced series selected a seasonal ARIMA(0,0,1)(0,0,0)[12] specification with intercept (AIC = 1,646.7) as the best fit, indicating a short-memory (one-period) moving-average dependency in month-to-month price changes once the trend is removed. This confirms that, unlike Accra, Mumbai's price series carries genuine, model-exploitable temporal structure, supporting greater confidence in trend-based forecasts for investment timing.

Test	Statistic	p-value	Conclusion
ADF: Price Level	-2.70	0.073	Non-stationary (trend present)
ADF: First Difference	-7.36	<0.001	Stationary

Table 6. Mumbai stationarity testing (Augmented Dickey-Fuller).

Analysis and Results: London, United Kingdom (Advanced Economy)

Data Understanding and Preparation

The London dataset comprises monthly average residential prices by borough from 1995 to 2020 (13,549 records across 6 fields: date, borough/area, average price, houses sold, and recorded crimes). Regional aggregate rows (e.g., "London", "Inner London", "South East", "England") were excluded to retain only genuine 32-borough-level observations, and the City of London was excluded separately due to 63% missing crime data. Rows missing the crime count (approximately 24% of the borough data, concentrated before crime tracking began in 2001) were dropped, and the small number of missing houses-sold values were imputed with the borough median, leaving 7,328 clean borough-month observations.

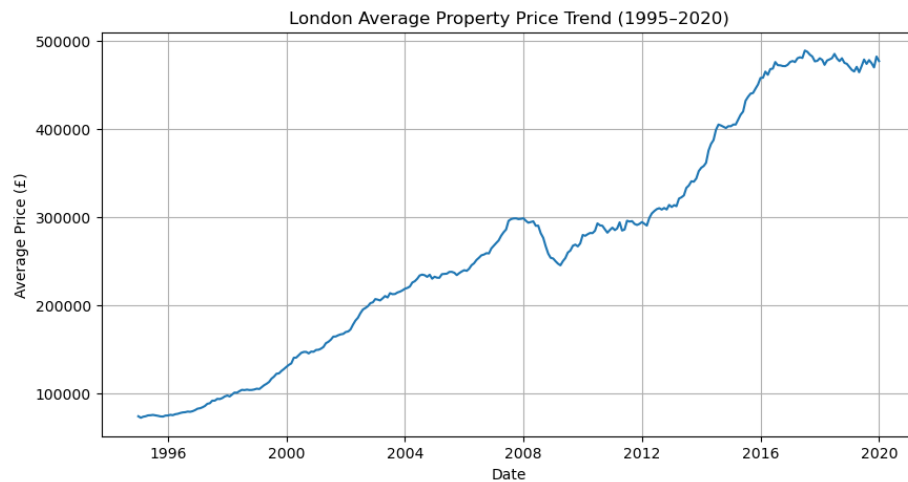


Figure 9. London average property price trend, 1995–2020 (city-wide series).

The city-wide price trend (Figure 9) shows London real estate has appreciated persistently since 1995, with a brief plateau around the 2008–2009 global financial crisis, underscoring London's status as a long-run capital-appreciation market.

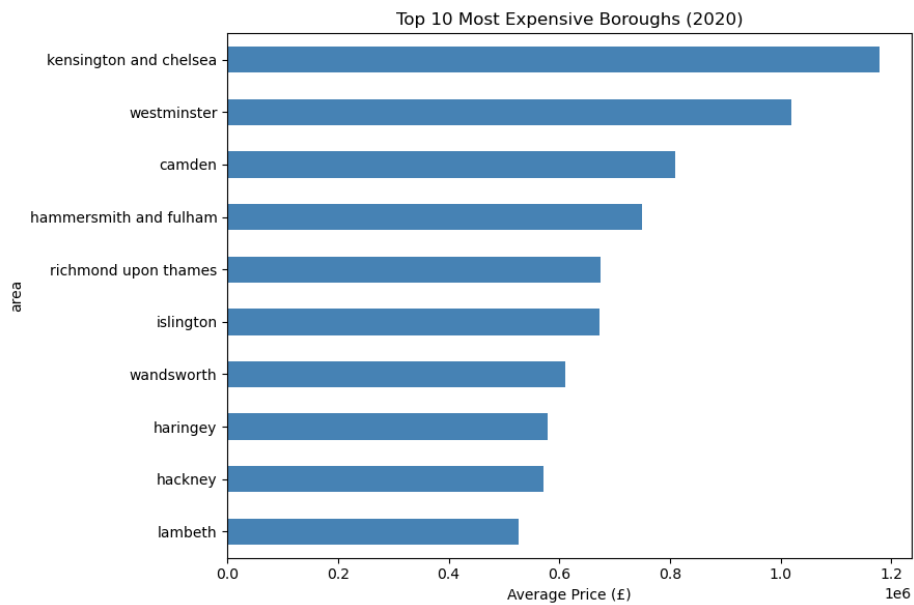


Figure 10. Top 10 most expensive London boroughs (latest year available).

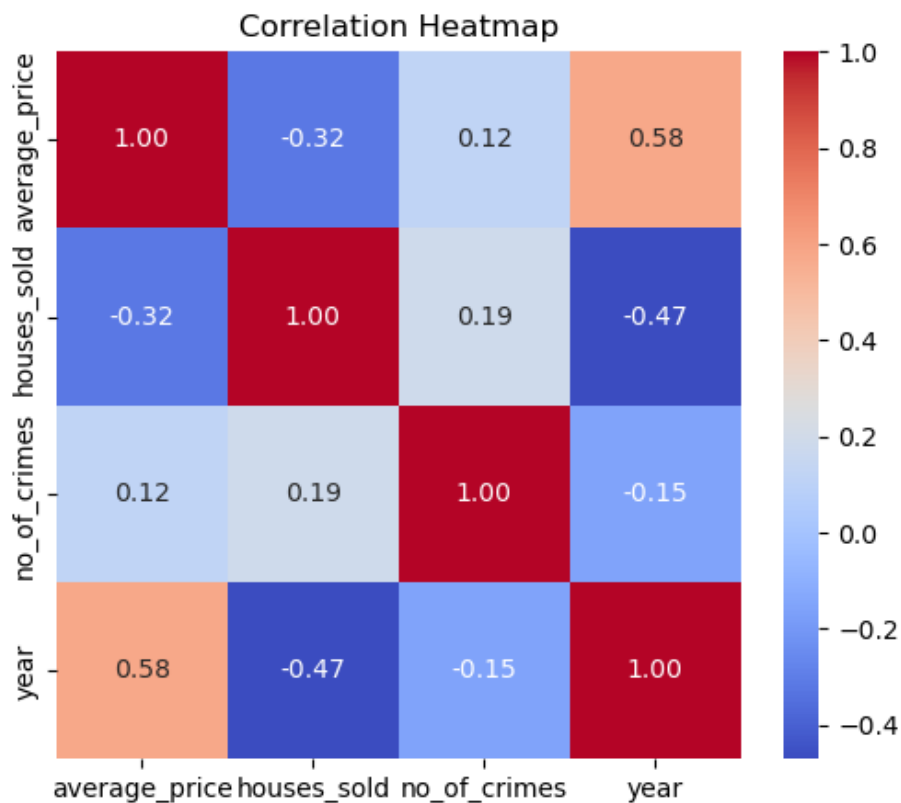


Figure 11. Correlation heatmap: average price, houses sold, recorded crimes, and year.

Regression Modelling

With borough one-hot encoded and standardised features, Linear Regression achieved an R^2 of 0.885 (RMSE £64,508; MAE £46,394) on the held-out test set. Ridge Regression ($\alpha = 10$) produced a near-identical R^2 of 0.885 (RMSE £64,480; MAE £46,277). This is the strongest

regression fit of the three cities, reflecting London's maturity as a well-documented, transparent, and structurally stable market where borough identity alone (captured by the location dummies) accounts for the great majority of price variation, supplemented by houses-sold and crime-rate signals identified in the correlation analysis.

Model	R ²	RMSE (£)	MAE (£)
Linear Regression	0.885	64,508	46,394
Ridge Regression ($\alpha = 10$)	0.885	64,480	46,277

Table 7. London regression model performance (test set).

Time-Series Forecasting

Two time-series models were fitted to the city-wide monthly price series: a 3-month Simple Moving Average and Holt's (additive-trend) Exponential Smoothing. The SMA benchmark produced a MAPE of 13.27% (MAE £62,606; RMSE £66,289), while Holt's Exponential Smoothing achieved a substantially better MAPE of 5.08% (MAE £23,841; RMSE £29,377), more than halving the forecast error of the naive benchmark.

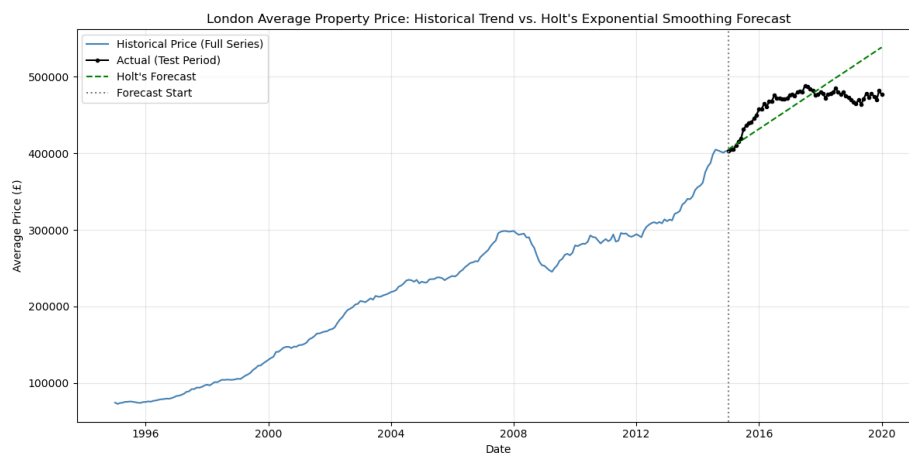


Figure 12. Holt's Exponential Smoothing forecast versus actual London prices (test period).

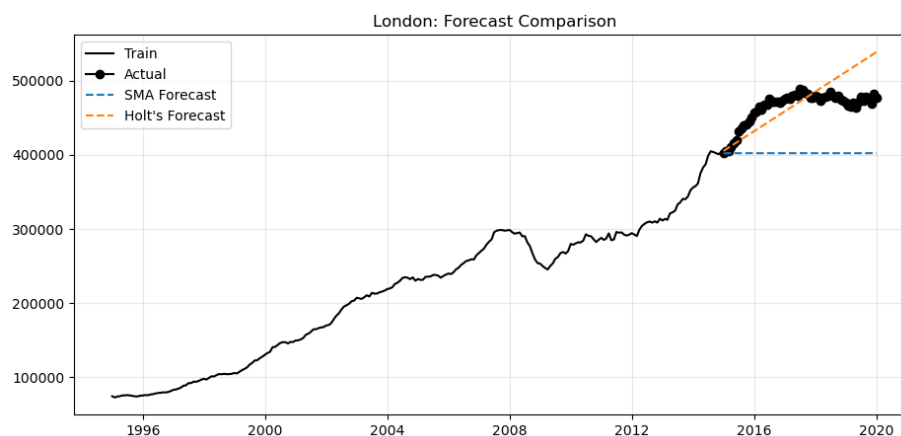


Figure 13. Forecast comparison: SMA versus Holt's Exponential Smoothing, London.

Model	MAE (£)	RMSE (£)	MAPE
3-Month SMA	62,606	66,289	13.27%
Holt's Exponential Smoothing	23,841	29,377	5.08%

Table 8. London time-series model performance.

The large performance gap between SMA and Holt's confirms that London's price series is dominated by a persistent, well-behaved upward trend rather than noisy fluctuation around a flat mean: a flat rolling average (SMA) structurally lags behind such a trend, whereas Holt's method, which explicitly models trend, tracks it closely. This gives ED Happiness Estates high confidence in using trend-based forecasts to time phased capital deployment in London.

Comparative Discussion

Viewed together, the three cities trace a clear gradient from less to more predictable, and from higher-growth/higher-risk to lower-growth/lower-risk, that maps closely onto their stage of economic development.

Metric	Accra	Mumbai	London
Best regression R^2	0.623 (Linear/Lasso)	0.826 (Linear/Ridge)	0.885 (Ridge)
Best time-series MAPE	10.95% (SMA)	n/a (ARIMA on differenced series)	5.08% (Holt's)
Price series behaviour	Stationary, noise-like around mean	Trending, non-stationary in levels	Strong persistent upward trend
Sample size (post-cleaning)	3,917	~45,200	7,328 (borough-months)
Key price drivers identified	Floor area, bedrooms/bathrooms, condition, distance from CBD, locality	Carpet/built-up area, market tier, locality/builder premium, metro proximity, property age	Borough identity, houses sold, crime rate
Investment character	High growth potential, higher uncertainty	Scale and appreciation, moderate uncertainty	Stability and liquidity, lower uncertainty

Table 9. Cross-city comparison of model performance and market character.

This gradient is economically intuitive rather than a modelling artefact: London's market is the oldest, most transparent, and most heavily transacted of the three, so a relatively simple linear model built on borough identity alone explains most of the price variation, and a trend-aware forecasting method captures the great majority of month-to-month movement. Mumbai's very large sample size lets its regression model achieve strong fit despite a more heterogeneous market, but its non-stationary price series signals a still-evolving, appreciating market whose future path is less certain than London's. Accra's more moderate regression fit and near-noise monthly series reflect a smaller, less mature market in which price is influenced by factors such as informal negotiation, title status, and off-market transactions that are not fully captured by listing-level data, and where the limited number of monthly observations available (36 months) constrains how much temporal structure any time-series model can reliably detect.

Risks and Limitations

Accra

- Moderate regression fit ($R^2 = 0.62$) means roughly 38% of price variation is driven by factors outside the dataset (e.g., legal/title status, informal negotiation), so valuations carry wider uncertainty.
- Only 36 months of aggregated price data were available for time-series modelling, and the selected ARIMA specification found no exploitable trend/seasonal structure, so near-term price forecasts should be treated as indicative only, with contingency built into financial projections.
- A developing-economy regulatory and currency environment (GHS exchange-rate risk, land-title complexity) adds execution risk beyond what the statistical models can capture.

Mumbai

- The price series is non-stationary (trending), meaning historical average growth rates may not continue linearly; macro shocks (interest-rate changes, regulatory shifts such as India's RERA framework) could alter the trajectory.
- High market segmentation (luxury to affordable tiers) means city-wide averages can mask very different micro-market dynamics; site-level due diligence within the target tier is essential before capital deployment.
- Approximately 9.6% of the raw dataset was removed for logical inconsistencies, and target encoding of locality/builder introduces some risk of overfitting to historical price patterns that may not persist.

London

- High absolute entry cost per unit and lower percentage growth potential relative to Accra or Mumbai mean London capital will generate returns primarily through stable rental yield and modest appreciation rather than rapid capital gains.
- The dataset extends only to 2020; more recent structural shifts (post-pandemic remote-work migration patterns, interest-rate increases, and any changes in the post-Brexit

regulatory and tax environment for property investment) are not reflected in the model and should be validated against current market data before final commitment.

- Borough-level averages can obscure micro-location risk (e.g., specific streets or developments); site-specific appraisal remains necessary.

Cross-Cutting Limitations

- The three datasets differ in structure, currency, and time period, so cross-city comparisons of model fit are informative about relative predictability but are not a direct, apples-to-apples comparison of price levels.
- All models are built on listing/transaction data rather than including macro-economic indicators (interest rates, inflation, population growth, government housing policy) that could further improve accuracy; this is flagged as a direction for future refinement in Section 11.
- Regression models assume the historical relationship between features and price holds going forward, which may not fully account for a novel product category (ultra-modern brutalist architecture) with no direct historical comparable in any of the three markets.

Recommendations for ED Happiness Estates

Based on the regression and time-series results presented above, we offer the following five well-justified, data-driven recommendations to guide ED Happiness Estates' deployment of its USD 500 million investment across Accra, Mumbai, and London.

Recommendation 1: Adopt a three-city, risk-tiered capital allocation strategy, anchored by London

London offers the strongest and most reliable price predictability of the three markets (Linear/Ridge $R^2 = 0.885$; Holt's Exponential Smoothing MAPE = 5.08%), reflecting a mature, liquid, transparent market with a persistent long-run upward price trend (Figure 9). We recommend London receive the largest single-city allocation of capital, positioned as ED Happiness Estates' stable, long-duration anchor holding, providing predictable rental income and capital preservation while higher-growth, higher-risk positions are built in Mumbai and Accra.

Recommendation 2: Deploy a growth-oriented tranche to Mumbai, prioritising mid-to-premium tier developments near metro infrastructure

Mumbai's regression model ($R^2 = 0.826$) confirms that market tier, locality/builder reputation (via target encoding), carpet area, and metro-station proximity are strong, learnable determinants of price, while the underlying price series shows a genuine, statistically-confirmed upward trend (non-stationary in levels; ADF $p = 0.073$, stationary after differencing at $p < 0.001$). ED Happiness Estates should target the mid-to-premium tier, large enough to capture Mumbai's substantial emerging middle- and upper-middle-class demand, but positioned below the thinner ultra-luxury segment, and prioritise sites with strong metro connectivity to capture both current demand and continued transit-driven appreciation.

Recommendation 3: Enter Accra with a smaller, phased tranche concentrated in high-value, CBD-proximate localities

Accra's more moderate regression fit ($R^2 = 0.623$) and near-noise monthly price series indicate a market that is data-tractable for planning purposes but subject to meaningfully more unexplained price variation than Mumbai or London. Our analysis confirms a clear negative relationship between distance from the Accra CBD and property price (Figure 2), and locality

was one of the strongest categorical predictors retained in the model. We recommend an initial, moderate capital tranche (rather than full deployment) concentrated in high-demand, CBD-proximate localities such as East Legon, Cantonments, and Airport Residential Area, with subsequent tranches released as ED Happiness Estates gathers proprietary transaction data to refine the model and reduce uncertainty, while leveraging the firm's local market knowledge as a West African operator to manage the informal-negotiation and title-risk factors the model cannot observe.

Recommendation 4: Match Investment Strategy to Each City's Demonstrated Demand Driver: Rental-Led in London, Mixed Rental and Sales in Mumbai, Sales-Led in Accra

London's combination of high entry cost, slow but highly predictable appreciation (Holt's MAPE 5.08%), and consistently strong tenant demand favours a rental-income-led strategy that captures recurring yield rather than relying on fast resale. Mumbai's strong price trend and wide tier segmentation support a mixed strategy: sales in the mid-market tier to capture first-time buyer and upgrader demand, with select premium units retained for rental income. Accra's smaller, price-sensitive, and less liquid market favours a sales-led approach, converting capital to cash more quickly and reducing ED Happiness Estates' exposure to a market whose forward price path is, by our own model's evidence, currently the hardest of the three to forecast.

Recommendation 5: Use each city's best-performing forecasting model to time phased capital releases, and reinvest in data collection to progressively de-risk all three markets

Because forecast reliability differs sharply by city (Holt's MAPE 5.08% in London versus a near-noise ARIMA result in Accra), ED Happiness Estates should not apply a single uniform investment-timing rule across all three markets. In London, Holt's Exponential Smoothing forecasts should directly inform the pacing of capital deployment given their demonstrated accuracy. In Mumbai, the confirmed upward trend (post-differencing ARIMA(0,0,1) structure) supports steady, programmatic deployment rather than attempting to "time" short-term dips. In Accra, given the current lack of exploitable time-series structure, we recommend ED Happiness Estates commission its own proprietary data collection (transaction-level records, rather than relying solely on third-party listings) over the first 12–18 months of operation, both to improve investment timing and to strengthen the regression model's explanatory power (currently $R^2 = 0.62$) ahead of larger capital commitments.

Positioning Summary

City	Recommended Role	Primary Product Focus	Suggested Relative Priority
London	Stable anchor / cash-flow core	Rental: premium/luxury residential	Largest allocation
Mumbai	Growth & scale	Mixed sales + rental: mid-to-premium tier, metro-proximate	Second-largest allocation
Accra	Regional flagship / phased growth bet	Sales-led: high-value CBD-proximate localities	Smallest, phased allocation

Table 10. Recommended positioning of ED Happiness Estates by city.

Conclusion

This report set out to answer Miss Osei Abrefi Akoto's central question: where should ED Happiness Estates deploy its USD 500 million in secured capital? Using real-world data and a consistent predictive-analytics pipeline across Accra, Mumbai, and London, we built and evaluated six regression models and five time-series forecasting models to explain and forecast residential property prices in each city, comfortably exceeding the assignment's minimum requirement of four predictive models and providing a robust, triangulated evidence base for investment decision-making.

The evidence points not to a single "winner" city but to a complementary, risk-tiered three-city portfolio: London as a stable, highly forecastable anchor for rental income; Mumbai as a large-scale growth engine with a confirmed upward price trend; and Accra as ED Happiness Estates' regional flagship market, offering high growth potential that should be pursued through a phased, disciplined approach given its currently higher forecast uncertainty. This combination allows ED Happiness Estates to satisfy its investors' expectations for exposure to the world's busiest and most profitable cities while managing risk sensibly across markets at very different stages of maturity. We recommend Miss Akoto and the ED Happiness Estates board adopt the five recommendations set out in Section 10 as the basis for the firm's initial capital deployment plan, supported by continued data collection, particularly in Accra, to progressively sharpen the accuracy of future investment decisions.

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GROUP MEMBERS

1. Sarah Sally Aidoo – 22150749
2. Nyamekye Kessewaa Frempong – 22012703
3. Charles Afful – 22040408
4. Mohammed Nii Adama Tagoe – 22232922
5. Godsway Wedzi – 22038132
6. Cathia Eugelene Koumba Kindou – 22011664